Accepted Manuscript online: 2025-01-14 Article published online: 2025-02-20

Distinct muscle growth and strength adaptations after preacher and incline biceps curls

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Keywords
exercise selection, muscle growth, resistance training, biceps brachii

received 27.08.2024
accepted after revision 14.01.2025
accepted manuscript online 14.01.2025
published online 20.02.2025

Bibliography
Int J Sports Med 2025; 46: 334–343
DOI 10.1055/a-2517-0509
ISSN 0172-4622
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Georg Thieme Verlag KG, Oswald-Hesse-Straße 50, 70469 Stuttgart, Germany

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Supplementary Material is available at https://doi.org/10.1055/a-2517-0509

ABSTRACT

We compared performing preacher and incline biceps curls on changes in elbow flexor muscle size and strength. This was a between-group repeated measure randomized trial. Sixty-three young women performed the preacher biceps curl ($n = 30$) or the incline biceps curl ($n = 33$) for 8 weeks, twice a week. We measured the muscle thickness of elbow flexors at the proximal, middle, and distal sites. We assessed the muscle strength using three repetition maximum tests in the preacher curl and the incline curl. We observed a greater increase in the proximal elbow flexor thickness in the incline biceps curl compared to the preacher biceps curl (mean difference = 0.08 cm [95 % confidence interval: 0.02, 0.13 cm]). We observed a greater increase in the distal elbow flexor thickness in the preacher biceps curl compared to the incline biceps curl (mean difference = 0.10 cm [95 %CI: 0.04, 0.15 cm]). The preacher biceps curl showed a greater increase in three repetition maximum tests in the preacher curl (mean difference = 1.88 kg [95 %CI: 1.14, 2.62 kg]). The incline biceps curl showed a greater increase in three repetition maximum tests in the incline biceps curl (mean difference = 0.86 kg [95 %CI: 0.10, 1.62 kg]). Our findings suggest regional differences in muscle growth induced by the preacher and incline biceps curls. Strength gains appear to follow the principle of specificity.

Introduction

Resistance training increases strength and muscle size [1]. When prescribing hypertrophy- and strength-oriented resistance training programs, exercise selection is a fundamental aspect because it appears to affect the magnitude of muscular adaptations [2, 3]. Nowadays, there is a great debate on – and some empirical support for – the potential superiority of specific resistance exercises to optimize gains in strength and muscle hypertrophy. For example, there are investigations demonstrating that a seated leg curl induces greater hypertrophy of hamstrings than the lying leg curl [4], straight-leg calf raise induces more favorable triceps surae growth than bent-leg calf raise [5], and elbow extension with an overhead arm position induces greater hypertrophy of the triceps brachii than elbow extension with a neutral arm position [6], although not universal [7]. Notably, this topic has been gaining momentum just recently. There is still controversy in the selection of exercises for op-

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timizing the growth of certain muscles, such as the elbow flexors [8, 9].

Common variations of resistance exercises targeting elbow flexor muscles involve manipulating the position of the shoulder (e.g., incline biceps curl and preacher biceps curl) [10]. The incline curl consists of performing the elbow flexion task with the shoulder in hyperextension, thus training in a more stretched muscle position, whereas the PC consists of performing the elbow flexion task with the shoulder in flexion, thus training in a more contracted muscle position. The only available study that compared head-to-head incline vs. preacher curl resistance exercises found that the preacher curl induced preferential distal hypertrophy, without differences in the proximal and middle sites [8]. Although not universal [11], this preferential distal elbow flexor muscle growth has been observed in different studies that investigated performing preacher dumbbell curls [12, 13]. Based on these observations, some could argue that the preacher dumbbell curl is a more effective resistance exercise as it induces greater distal muscle growth and similar hypertrophy at the other sites.

Of note, another report observed preferential proximal muscle growth by combining the incline curl, biceps curl (neutral shoulder position), and preacher curl compared to performing exclusively biceps curl [9]. The authors attributed proximal site hypertrophy to the incline curl exercise [9]. Of note, since the combination of different exercises in the mentioned study [9], it is not possible to be assertive whether this was in fact the case. On muscular strength, less is known. The only study, by Zabaleta-Korta et al. [8], that compared the two exercises head-to-head did not measure this outcome. Thus, a head-to-head comparison of the two exercises with measurements of proximo-distal muscle size changes and strength is necessary to elucidate this topic. In the present study, we compared the effects of preacher vs. incline curls on the elbow flexor muscle size and strength. We hypothesized that the preacher curl could induce greater distal muscle hypertrophy, but without differences in other sites, and that strength adaptations would be greater in the trained exercise, in accordance with the principle of specificity.

Material and Methods

Study design

This was a between-group repeated measure randomized clinical trial. Untrained young women were randomly allocated to one of two possible training groups: the preacher curl (PC) or the incline curl (IC) exercise. Before the training intervention, participants performed two familiarization training sessions interspersed by 48 h. Seventy-two hours after the final familiarization session, the muscle thickness of elbow flexors was acquired at proximal, middle, and distal sites. Twenty-four to 48 hours later, three repetition maximum (3RM) tests were performed on preacher and incline curl exercises. The training period was initiated, with groups performing their respective resistance exercises. The resistance training program was performed two times a week, for 8 weeks, totaling 16 training sessions. The weight was progressively increased in the two training groups throughout the 8 weeks. Reassessments of the muscle thickness and 3RM were performed at an interval of 72–120 h after the final training session (session 16).

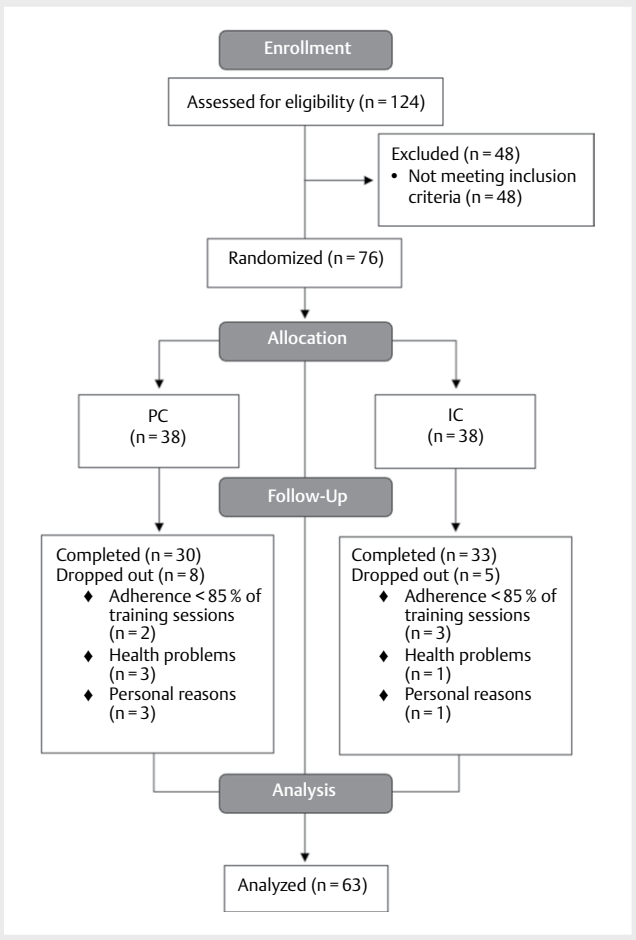
Subjects

Seventy-six untrained young women were recruited from a university population to participate in this study. Potential participants were contacted through dissemination on social media and folders distributed in places of greater circulation within the campus of the local University. Individuals underwent a clinical anamnesis and answered the physical activity readiness questionnaire (PAR-Q). The inclusion criteria were not responding "yes" to one or more PAR-Q questions, being between the ages of 18 and 35 years, not having myoarticular problems that could compromise the performance of the training protocol, not being a user of dietary supplements and/or anabolic steroids (self-reported information), and not being engaged in resistance training for at least 3 months before starting the participation. Participants who did not attend at least 85 % of training sessions and/or missed three consecutive training sessions were excluded from the analyses. Sixty-three young women (PC, $n = 30$ [22.4 ± 3.1 y old, 63.9 ± 11.7 kg, and 163.8 ± 5.3 cm] and IC, $n = 33$ [23.1 ± 3.8 y old, 66.7 ± 15.5 kg, and 162.6 ± 6.5 cm]) ultimately completed the study and were included for final analyses. ▶ **Fig. 1** illustrates the sampling process, with details on the number of participants recruited and interviewed, allocation to groups, dropouts throughout the study, and the final number of participants who completed the assessments and training protocol and were included in the analyses. All participants were instructed to maintain their dietary intake and routines during the participation. This study was conducted according to the Declaration of Helsinki and the research project was approved by the Institution's Ethics Committee (approval #CAEE: 29064620.2.0000.5231). This investigation meets the guidelines set forth by the International Journal of Sports Medicine [14]. All participants were informed of the inherent risks and benefits signing an informed consent form.

Sample size justification

The target sample size for this study was 60 participants; however, to account for a dropout rate of ~20 %, we aimed to recruit ~75 participants. Sample size determination for the study was made based on previous research on the topic with similar study design (e.g., between-subject) and pragmatic considerations (i.e., resources constraints and time available). Specifically, this aimed sample size was greater than similar between-subject pre–post studies investigating the influence of exercise selection on muscle hypertrophy. Based on previous research projects conducted in our laboratory, time, and associated costs to complete the data collection for the current study, we realized that recruiting more than 75 participants was not feasible due to resource constraints. Additionally, we conducted a sensitivity analysis using G*Power software (version 3.1.9.6) to identify what effect we were able to detect given the number of observations collected in total ($n = 63$; alpha level of 5 %; and desire 80 % power). The sensitivity power analysis reveals that the study has 80 % power to detect effects of at least Cohen's $f = 0.3$ (Cohen's $d = 0.6$). Compared to previous training studies with untrained participants [9, 15], it is very likely sufficient to capture effect sizes for within-group changes ($d = 0.7$ to 1.0) and potentially sufficient to detect between-group differences in muscle growth ($d = 0.4$ to 0.6). The critical effect size is likely sufficient to detect

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▶ **Fig. 1** CONSORT flow chart [36]. IC, incline curl; PC, preacher curl.

between-group differences in 3RM strength outcomes [2, 16]. Also, the adopted statistical model (i.e., analysis of covariance; ANCOVA) has more power [17–20], particularly when pre-training and outcome measures (e.g., post-training or change score) are correlated [17, 18, 20], as typically is the case for muscle size and strength outcomes. Thus, covariation for the baseline may enhance the detection of treatment effects in randomized clinical trials.

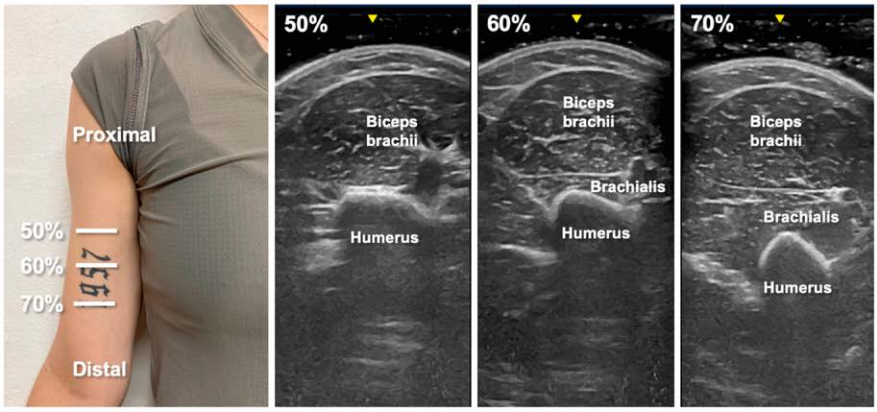
Muscle thickness assessments

The muscle thickness assessment of elbow flexors was taken via B-mode ultrasound imaging (FIGLABS, FP-102, SAEVO, São Paulo, Brazil), with a 7.5-MHz linear probe (model L741; SAEVO). On arrival at the laboratory, the participant had the right arm measured and marked. Reference lines were drawn on the participant's skin with a dermatographic pen on the sites at which images were obtained. Images of the anterior arm (biceps brachii and brachialis muscles) were obtained at 50 % (proximal), 60 % (middle), and 70 % (distal) of the distance between the acromial process of the scapula and the lateral epicondyle of the humerus. The arm length was recorded and used for the post-training assessment. After being marked, the participant was positioned supine on a stretcher for 10 minutes before the start of the measurement. Although the time participants spent lying down did not appear to affect muscle thickness measurements [21], we chose to standardize that all participants spent this amount of time lying down before scanning. A generous quantity of water-soluble transmission gel was applied over the muscle being assessed, without compressing the skin. The elbow flexor muscle thickness was defined as the distance from the superficial interface, corresponding to the aponeurosis of the subcutaneous adipose tissue, and the edge of the bone tissue. Images were obtained with the probe held horizontally to produce a transverse plane image (top view) of the underlying tissues. Two assessors participated in measurement procedures: the first handled the probe and the second was responsible for freezing the images. When the quality of the image was deemed to be satisfactory, the second assessor (blinded to group allocation) obtained muscle thickness dimensions. The muscle thickness dimensions were obtained using the ultrasound machine's calculation package. Ultrasound settings (frequency, 7 MHz; gain, 70 dB; depth, 8.0 cm) were recorded and kept consistent across the study. During pre-training assessments, 15 participants were randomly chosen to be evaluated on 2 days, separated by 72 hours, to determine the reliability of the muscle thickness measurements. We performed two independent collections followed by quantifications with these 15 participants. The second image was taken side-by-side with the previous image taken on day 1, allowing the assessor to identify landmarks within the muscle to ensure a similar measurement site was used. The image taken on day 1 was cropped to remove the depth to prevent influencing the second measurement. This procedure was repeated in the post-training assessment. The intraclass correlation coefficients (ICC3,1) for the elbow flexors at proximal, middle, and distal sites were 0.968, 0.970, and 0.996; the coefficients of variation were 2.67, 2.19, and 1.48 %; the standard errors of measurement (SEMs) were 0.064, 0.055, and 0.057 cm, respectively. Participants' skin was marked, and typical proximal, middle, and distal ultrasound images of the anterior arm are displayed in ▶ **Fig. 2**.

Three repetition maximum (3RM) test

Muscle strength was assessed by applying 3RM tests in the preacher curl and incline curl exercises. Both exercises were performed with dumbbells and with supinated hands. Details on the apparatus and participant's position during both resistance exercises are described in the following section. Three days of 3RM testing were performed on alternate days, 48 hours apart. Each participant performed three attempts per testing day for both exercises. In each testing session, the participants initially executed a specific warm-up composed of one set of 6–10 repetitions with an intensity of approximately 50 % of the weight to be used in the first attempt of the 3RM test in the specific exercise. The first attempt was initiated after a passive rest of 2 minutes after the end of the warm-up. During each attempt, strong verbal encouragement was provided throughout the test. In all trials, participants were instructed to perform four repetitions. When four repetitions were performed successfully with the proper form, the weight was increased by 2–5 % for the next attempt. When the participant could not even perform three complete repetitions, the weight was reduced by the same magnitude. A rest interval of 3–5 minutes was granted between attempts and 30 min between the preacher and inline curl exercises. The higher weight lifted with the proper technique

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▶ **Fig. 2** Participants' skin marked, and typical proximal, middle, and distal ultrasound images of the anterior arm.

among the attempts for each exercise was recorded and used in the statistical analyses. Professionals with experience in 3RM tests supervised all testing sessions. The same evaluators supervised pre and post-training tests. The reliability of the 3RM in preacher biceps curl (3RM-PC) and incline curl (3RM-IC) was determined from the values obtained on the second and third days of 3RM testing. The ICC3,1 values for the 3RM-PC and 3RM-IC were 0.959 and 0.978; the coefficients of variation were 4.03 and 2.45 %; the SEMs were 0.39 and 0.34 kg, respectively.

Resistance training program

A 1-week familiarization period was provided before the study whereby participants performed preacher and incline curl exercises over 2 non-consecutive days (the preacher curl exercise in the first and incline curl in the second familiarization training session). Participants performed both resistance exercises using a minimal weight for two sets of 12 repetitions and far from task failure. This theoretically promoted repeated bout effects and thus minimized unwanted muscle soreness from interfering with training. After the familiarization and pretraining measurements, the resistance training program was initiated. The resistance training program was performed two times per week in the afternoon for 8 weeks. Resting between sessions typically ranged from 48 to 72 hours within a week. The PC was performed on a bench consisting of a 30 ° inclination. During the PC exercise, the movement was performed with the participant in a standing position, with the upper arms positioned on the preacher curl bench with a ~45 ° shoulder flexion. The preacher curl movement was started with the elbow fully extended, going to ~120 ° of flexion and returning to the starting position. The incline curl was performed on a bench with a 45 ° inclination. During the incline curl, the movement was performed with the participant in a seated position and with ~30 ° shoulder hyperextension. The participants started with the elbow fully extended with arms perpendicular to the ground. Then had to perform the elbow flexion until ~120 ° during the concentric muscle action, followed by the controlled eccentric muscle action. Participants in both exercises were requested to keep their hands supinated. Representative photographs of the initial and final phases of both exercises are presented in ▶ **Fig. 3**. Both exercises were performed bilaterally with dumbbells. Smaller possible weight adjustments were 0.2 kg. The training protocol consisted of three sets of 8–12 repetitions, executed until momentary concentric failure (i.e., when the participant has reached the point where, despite trying to do so, they cannot complete the concentric muscle action of



▶ **Fig. 3** Initial and final positions of preacher (**A**) and incline (**B**) biceps curls.

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their current repetition). Whenever the subject reached 12 repetitions in a given set and reported that they could perform 1 or more repetitions, the weight was increased by 2–5 % for the next set to ensure that the participants kept performing the sets to (or very near to) failure in the established repetition range. A 2-minute rest interval was given between sets. Resistance exercises were performed at a tempo of 1/0/2/0 (concentric, concentric–eccentric transition, eccentric, and eccentric–concentric transition phases, respectively). Participants performed all training sessions under specialized supervision (1:2, participant:supervisor ratio) to ensure the proper technique and safety. All weights and repetitions performed were recorded. The volume load was calculated as the number of sets *x* number of repetitions *x* weight lifted. The volume load of the first training session in week 1 was used as each participant's reference value. Thereafter, volume load progression was calculated based on percentage differences between the average volume load produced between training sessions 1 and 2–16. In addition, the total training volume (TTV) was calculated for each subject as the sum of the training volume produced in each resistance training session throughout the training period (sessions 1–16). The volume load progression and TTV were calculated. Of note, the current experiment presented herein was part of a larger training program that also included a lower body training protocol (lower body data reported elsewhere). The biceps curl exercise (preacher or inclined biceps curl) was the second exercise of each training session.

Dietary intake

A nutritionist instructed the participants to perform the food record on 3 non-consecutive days, 2 weekdays (typically Tuesday and Thursday) and 1 day on the weekend (typically Saturday), in the first and last training weeks. All participants received instructions for filling out the records using household measures. The intake of carbohydrates, lipids, proteins, and total energy intake was processed using nutritional analysis software (Avanutri Processor Nutrition, v. 3.1.0, Rio de Janeiro, RJ, Brazil). All participants were instructed to maintain their usual food intake throughout the study. The values of total energy intake and macronutrients used in the analyses were calculated based on the arithmetic mean of the 3-day food records.

Statistical analysis

Data distribution and variance homogeneity were verified through visual inspection of Q–Q plots, Shapiro–Wilk's tests, and Levene's tests. Two-way analysis of variance (ANOVA) was used to compare the dietary intake between groups in the first and last weeks of intervention. The comparison of the effects of the PC and IC on the elbow flexor muscle thickness and 3RM was made using an analysis of covariance (ANCOVA) of the raw difference between pre- and post-training measures with the pretraining score as a covariate. The interpretation of data was based on 95 % confidence intervals (95 %CI) of the change score (i.e., there was a significant difference when the inferior and superior confidence limits did not cross zero) [22]. The *p* values for within- (adjusted marginal means of the change score compared to zero) and between-group comparisons were also presented. We present within- and between-group results as the mean differences ($mean_{diff}$) and associated 95 %CI. The effect size (ES), commonly known as Cohen's *d*, was calculated as the post-training group mean minus the pre-training mean, divided by the group-pooled pretraining standard deviation [23]. An unpaired *t*-test was used to compare training attendance and TTV between the groups. The volume load progression slopes were generated using linear regression and compared through an *F*-test. For all statistical analyses, significance was accepted at $p < 0.05$. The data were stored and analyzed using JASP software (version 0.14.1, Amsterdam, NL). The data are presented as mean and standard deviations. We used GraphPad software (version 10.4.0, Boston, USA) and the Estimation Statistics site (https://www.estimationstats.com) to create the figures for the main outcomes (i.e., muscle thickness, 3RM strength, and TTV).

Results

Dietary intake and training attendance

There were no main effects of time, group, or time and group interactions for carbohydrates, proteins, lipids, and total energy intake ($p \geq 0.346$; see Supplementary Digital Content S1 [available in the online version only], Dietary intake for details). Session attendance was 95.6 ± 4.9 % for the PC and that was 95.8 ± 4.3 % for the IC, with no difference between the groups ($p = 0.859$).

Muscle thickness

Pre- and post-training values of the elbow flexor muscle thickness at different sites are displayed in ▶ **Table 1**. For proximal elbow flexors, both groups increased their muscle thickness compared to pretraining ($p < 0.001$). A significant group effect was observed for the changes in the elbow flexor muscle thickness at the proximal site ($F = 7.392$; $p = 0.009$). The mean absolute change was significantly greater in the IC compared with the PC ($mean_{diff} = 0.08$ cm [95 %CI: 0.02, 0.13 cm]). Relative changes from pretraining and ES for the elbow flexor muscle thickness at the proximal site were as follows: PC = + 8.3 %, ES = 0.43; IC = + 11.2 %, ES = 0.60. Within-group $mean_{diff}$ and 95 %CI are depicted in ▶ **Fig. 4A**.

For middle elbow flexors, both groups increased their muscle thickness compared to pretraining ($p < 0.001$). No significant group effect was observed for the changes in the elbow flexor muscle thickness at the middle site ($F = 0.083$; $p = 0.774$). Between-group $mean_{diff}$ and 95 %CI were as follows: − 0.01 cm (− 0.07, 0.05 cm). Relative changes from pretraining and ES for the elbow flexor muscle thickness at the middle site were as follows: PC = + 9.3 %, ES = 0.60; IC = + 8.6 %, ES = 0.57. Within-group $mean_{diff}$ and 95 %CI are depicted in ▶ **Fig. 4B**.

For distal elbow flexors, both groups increased their muscle thickness compared to pretraining ($p < 0.001$). A significant group effect was observed for the changes in the elbow flexor muscle thickness at the distal site ($F = 11.637$; $p = 0.001$). The mean absolute change was significantly greater in PC compared with IC ($mean_{diff} = 0.10$ cm [95 %CI: 0.04, 0.15 cm]). Relative changes from pretraining and ES for the elbow flexor muscle thickness at the distal site were as follows: PC = + 10.3 %, ES = 0.79; IC = + 6.4 %, ES = 0.50. Within-group $mean_{diff}$ and 95 %CI are depicted in ▶ **Fig. 4C**.

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▶ **Table 1** The muscle thickness and 3RM values at pre- and post-training, and within group $mean_{diff}$ with 95 %CI[A].

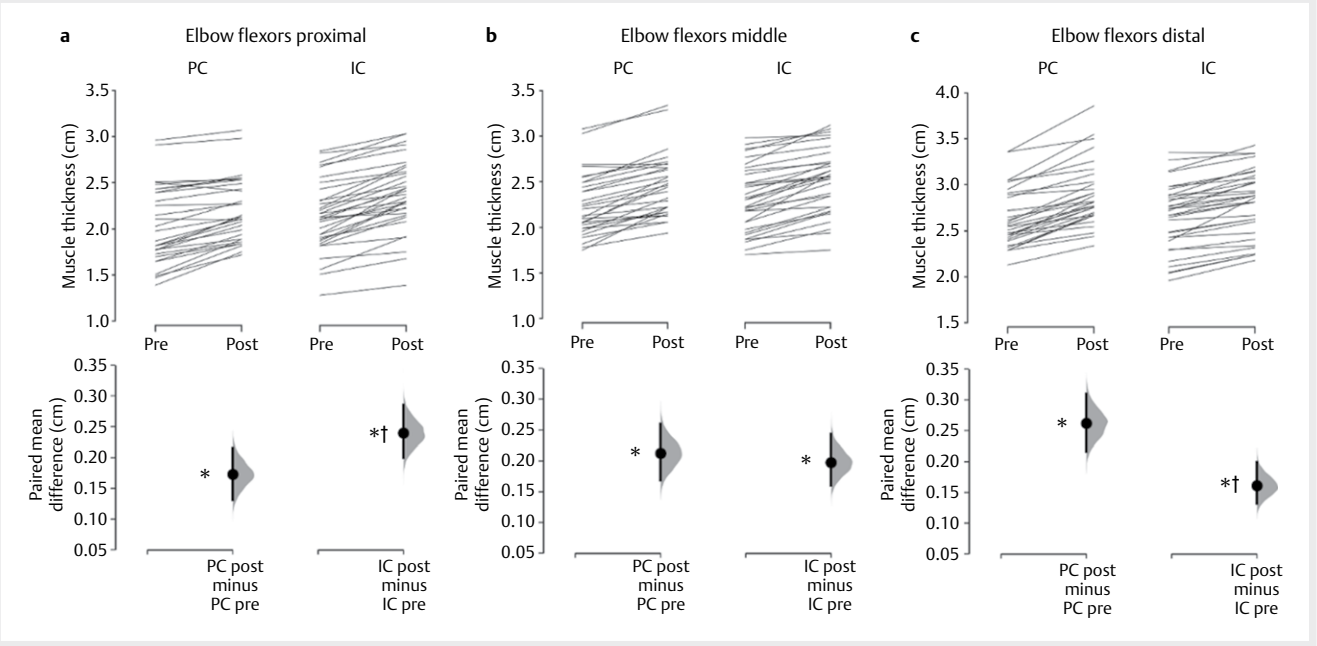
Variables	PC (n = 30)			IC (n = 33)		
	Pre	Post	Δ (95 %CI)	Pre	Post	Δ (95 %CI)
Elbow flexor proximal (cm)	2.04 ± 0.42	2.21 ± 0.35	0.16 (0.11, 0.21)*	2.14 ± 0.38	2.38 ± 0.39	0.24 (0.19, 0.29)*†
Elbow flexor middle (cm)	2.25 ± 0.34	2.46 ± 0.34	0.21 (0.15, 0.26)*	2.32 ± 0.36	2.52 ± 0.35	0.20 (0.15, 0.25)*
Elbow flexor distal (cm)	2.63 ± 0.32	2.90 ± 0.35	0.26 (0.21, 0.31)*	2.65 ± 0.36	2.82 ± 0.36	0.16 (0.11, 0.20)*†
3RM-PC (kg)	11.50 ± 2.94	17.42 ± 3.33	5.91 (5.29, 6.52)*	12.02 ± 2.50	15.94 ± 2.64	4.02 (3.44, 4.61)*†
3RM-IC (kg)	11.42 ± 2.32	14.13 ± 2.90	2.70 (2.07, 3.34)*	11.37 ± 1.71	14.85 ± 2.19	3.56 (2.96, 4.17)*†

Abbreviations: ANCOVA, analysis of covariance; CI, confidence interval; IC, incline curl; PC, preacher curl; 3RM-IC, three repetitions maximum in the incline curl; 3RM-PC, three repetitions maximum in preacher curl.

Notes: Pre and post-training data are presented as mean and standard deviation, while within group $mean_{diff}$ as mean and 95 %CI.

*p < 0.05 vs. pre; †p < 0.05 vs. PC.

[A]Within group $mean_{diff}$ and 95 %CI values were determined using an ANCOVA between-group test with the pretraining score as the covariate.



▶ **Fig. 4** The paired mean differences for within-group comparisons are shown in the estimation plots. For (**A**) (elbow flexor proximal), (**B**) (elbow flexor middle), and (**C**) (elbow flexor distal) panels, the raw data are plotted on the upper axes; each paired set of observations is connected by a line. Each paired mean difference is plotted as a bootstrap sampling distribution on the lower axis. Mean differences are depicted as dots; 95 %CI values are indicated by the ends of the vertical error bars. CI, confidence interval; IC, incline curl; PC, preacher curl; *p < 0.05 vs. pre; †p < 0.05 vs. PC.

3RM

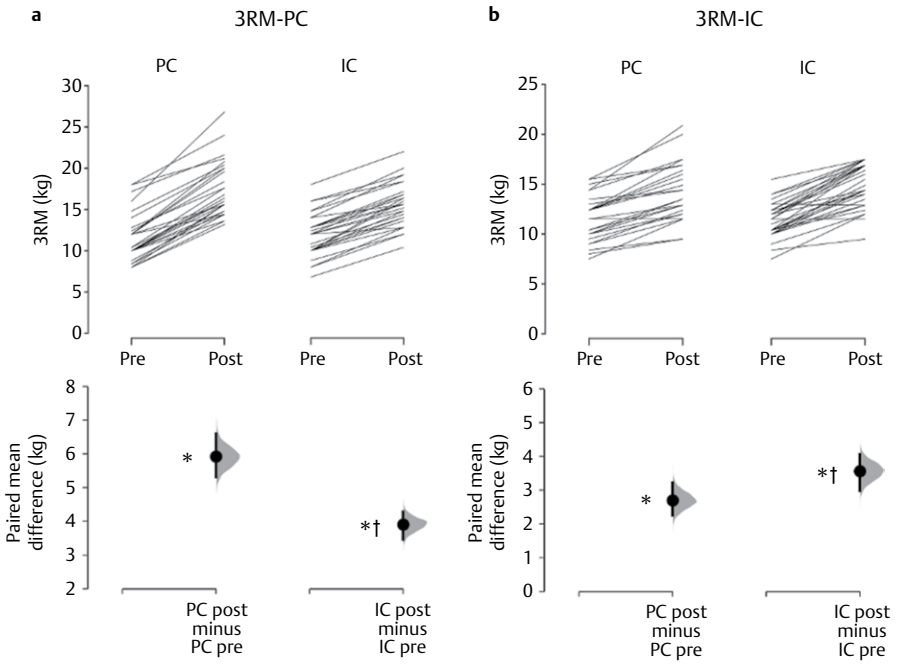
Pre- and post-training values of 3RM-PC and 3RM-IC are displayed in ▶ **Table 1**. For the 3RM-PC, both groups increased their muscle strength compared to pretraining (p < 0.001). A significant group effect was observed for the changes in the 3RM-PC (F = 25.924; p < 0.001). The mean absolute change was significantly greater in the PC compared with the IC ($mean_{diff}$ = 1.88 kg [95 %CI: 1.14, 2.62 kg]). Relative changes from pretraining and ES for the 3RM-PC were as follows: PC = + 51.5 %, ES = 2.18; IC = + 35.1 %, ES = 1.44. Within-group $mean_{diff}$ and 95 %CI are depicted in ▶ **Fig. 5A**.

For the 3RM-IC, both groups increased their muscle strength compared to pretraining (p < 0.001). A significant group effect was observed for the changes in the 3RM-IC (F = 5.129; p = 0.027). The mean absolute change was significantly greater in the IC compared with the PC ($mean_{diff}$ = 0.86 kg [95 %CI: 0.10, 1.62 kg]). Relative changes from pretraining and ES for 3RM-IC were as follows: PC = + 23.7 %, ES = 1.34; IC = + 30.6 %, ES = 1.73. Within-group $mean_{diff}$ and 95 %CI are depicted in ▶ **Fig. 5B**.

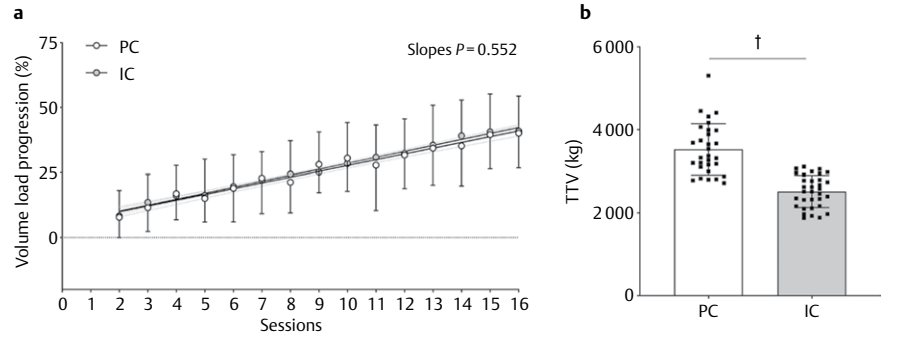
Volume load progression and TTV

Group analyses revealed that the progression of the volume load throughout the resistance training program was similar between the PC and the IC. The PC produced greater TTV than the IC (3527.0 ± 617.2 kg vs. 2516.0 ± 384.6 kg; p < 0.001). ▶ **Fig. 6** shows

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▶ **Fig. 5** The paired mean differences for within-group comparisons are shown in the estimation plots. For (**A**) (3RM-PC) and (**B**) (3RM-IC) panels, the raw data are plotted on the upper axes; each paired set of observations is connected by a line. Each paired mean difference is plotted as a bootstrap sampling distribution on the lower axis. Mean differences are depicted as dots; 95 %CI values are indicated by the ends of the vertical error bars. CI, confidence interval; IC, incline curl; PC, preacher curl; 3RM-PC, three repetition maximum in the preacher curl; 3RM-IC, three repetition maximum in the incline curl; *$p < 0.05$ vs. pre; †$p < 0.05$ vs. PC.



▶ **Fig. 6** Panel (**A**) shows the volume load progression per session for the PC and IC groups, with slopes (continuous straight lines) and 95 % confidence intervals (dotted lines). Panel (**B**) shows the TTV following the intervention. IC, incline curl group; PC, preacher curl group; TTV, total training volume; †$p < 0.05$ vs. PC.

the volume load progression from session to session and the TTV following intervention.

Discussion

We compared the effects of two exercises that target the elbow flexors on muscle development and strength changes in untrained young women. Our main findings were (a) the incline curl induced the greater proximal elbow flexor muscle thickness increase than the preacher curl; conversely, (b) the preacher curl elicited the greater distal elbow flexor muscle thickness increase than the incline curl; (c) the group that performed the preacher curl exercise increased more 3RM-PC; and (vi) the group who performed the incline curl exercise increased more 3RM-IC. Our hypothesis of greater hypertrophy of the distal elbow flexors in response to the preacher curl exercise was supported by our findings; however, the hypothesis of similar hypertrophy in the proximal and middle sites was only partially supported, as the incline curl exercise induced

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greater proximal elbow flexor hypertrophy. Our hypothesis on the specificity of strength adaptations was supported by our findings. In the following paragraphs, we contextualize our findings, proposing potential explanations and practical applications, as well as highlighting the strengths and limitations of the present study.

In the present study, we observed slightly greater proximal elbow flexor hypertrophy after incline curls. Our findings differ from a previous study that compared head-to-head incline and preacher curls and observed no differences between the exercises in the growth of the proximal site [8]. Of note, another study reported preferential hypertrophy of the proximal elbow flexors when combining the incline curl, arm (neutral) curl, and preacher curl [9], illustrating an inconclusive scenario. Although it is not possible to determine exactly the reasons for the divergence between the previous study [8] and ours, it is possible that this may be related to the sample size (larger in the present study) and/or training status (untrained women in the current study and resistance-trained women in the previous one [8]). We observed relatively small between-group differences, so such an effect is probably detectable with larger sample sizes. Also, it has been suggested that untrained individuals experience greater changes than resistance-trained individuals [24]. Taken together, these factors may help to contextualize the divergent findings between our study and the previous one [8]. Interestingly, in another study that combined incline, neutral, and preacher curls, the authors attributed preferential muscle growth to the incline curl in untrained individuals [9]. However, as different exercises were combined, this suggestion remained a hypothesis. Of note, our results support the suggestion that the incline curl may induce preferential proximal elbow flexor muscle growth.

Our finding of greater proximal muscle growth in the incline curl than the preacher curl may be explained, at least in part, by the between-exercise differences in the trained muscle length and the portion of the range of motion in which there is greater external torque. During the incline curl, the shoulder is placed in a hyperextended position [8]. As the biceps brachii cross the shoulder joint, this muscle is placed in a more elongated length, especially the most proximal sites. Parallel to this, during the incline curl, the greater external torque in this movement occurs in the final half of the range of motion (~70–120° of elbow flexion) [8]. In contrast, during the preacher curl, the shoulder is placed in a flexed position and there is less external torque in the final half of the range of motion [8]. Based on these aspects, we speculate that a combination of stretching at the proximal muscle site and higher external torque at the final half of the range of motion during the incline curl may have elicited greater mechanical tension at this muscle (ie., biceps brachii) or proximal muscle site compared to the preacher curl. Collectively, these factors may explain, at least in part, the greater proximal muscle growth in response to the incline curl exercise. Interestingly, there is modeling and empirical work suggesting non-uniform strain and shortening in the biceps brachii [25, 26], supporting the notion that specific sites of the biceps brachii may experience differential stimuli during the elbow flexion task. Notably, in our study, we observed a between-group difference of trivial magnitude. Moreover, discrepant results were observed in previous studies [8, 9]. Thus, replication studies are needed.

We observed greater distal hypertrophy of the elbow flexors after the preacher curl exercise. Our findings corroborate a previous study that compared incline vs. preacher curls and reported more favorable distal hypertrophy after the preacher curl [8]. This distal hypertrophy of the elbow flexors in response to the preacher curl has also been observed in other investigations that compare the effect of the range of motion [12, 13]. During the preacher curl with free weights (e.g., dumbbells), the greatest external moment arm occurs in the initial portion of the range of motion (~0–70° of elbow flexion) [8]. On the other hand, in the incline curl, the external torque at the beginning of the range of motion is minimal. Thus, it is reasonable to suggest that the greater distal hypertrophy of the elbow flexors after the preacher curl may be related, at least in part, to the higher external torque placed at the beginning of the movement, which potentially induced a high strain on a distal muscle site relative to the incline curl exercise. Interestingly, when the preacher curl is performed on equipment with a cable and cam system (cam shape was not informed), elbow flexor muscle hypertrophy appears to occur homogeneously across the three sites of the elbow flexors [11]. In equipment with a cam system, particularly with a constant radius, the external moment arm varies less [27, 28]. Taken together, such observations allow us to speculate that the point of the greatest external torque may influence the magnitude and site where greater hypertrophy will occur. Of note, this hypothesis needs investigations with a specific design to test it [29].

Regarding muscle strength adaptations, we observed greater strength gains in the preacher curl exercise after preacher curl training and greater strength gains in the incline curl exercise after incline curl training. Our findings are in accordance with the principle of specificity. Given that both groups trained at the same intensity, it is likely that adaptations such as an increased agonist voluntary activation have occurred at the same magnitude. One possible factor that may help to explain the specificity of the strength gains may be the subtle but sufficient differences in demand for intermuscular coordination. For example, during the incline curl, given that there was no arm support, probably there was a greater need to stabilize the shoulder joint compared to the preacher curl. The long head of the triceps brachii – antagonist in the elbow flexion movement – acts to hold the head of the humerus in the glenoid cavity [30]. On the other hand, during the preacher curl, the demand to stabilize the shoulder joint, and therefore the demand on the antagonist muscles, is likely lower. Such differences may have impacted the coactivation of antagonist muscles, contributing, in part, to task-specific improvements in coordination and, as a consequence, in strength adaptations. Parallel to this, differences in motor task refinement and the impact on individual self-efficacy also likely help to explain the specificity of strength adaptations [31, 32]. Although reasonable, studies that investigate mechanisms that help to explain the specificity of strength adaptations when comparing resistance exercises are necessary.

Our study has strengths and limitations that deserve to be mentioned. Participants were supervised throughout the training program. This allowed participants to be encouraged to apply high effort in all sets and go to task failure in all sets while maintaining the proper technique. Monitoring and observing no differences in eating habits increases our confidence that changes between groups were not influenced by this factor. As a limitation, in the present

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study, we measured the muscle thickness of the elbow flexors, without differentiating between biceps brachii and brachialis. There is rationale to suggest that the contribution of each may differ (e.g., the point in the range of motion where there is the greatest internal moment arm appears to be different between biceps brachii and brachialis [33]). Therefore, future studies may consider measuring these muscles individually. Another limitation is, perhaps, the lack of the measurement of physical activity levels. Thus, it remains unknown whether or not this factor may have affected our findings. Also, some may argue that the duration of the study was short. This could explain the small differences in muscle growth between the groups. Of note, the greatest rates of muscle growth appear to occur in the initial months of training [34]. Therefore, we believe that the duration of the present study was sufficient and perhaps the differences are indeed small. Also, a delimitation of the present study is the absence of measures of muscle activation/excitation (e.g., T2 relaxation time and electromyographic signal amplitude) in parallel with the muscle size measures. Future studies should consider the inclusion of proxies of muscle participation; this will provide insights into the adaptation process.

Our results on regional muscle hypertrophy may be of interest from an aesthetic point of view (e.g., bodybuilding competitors). Of note, regional muscle growth may be a mediator for increased strength (e.g., greater proximal or distal muscle growth may allow overcoming the sticking point in a given resistance exercise), but this hypothesis lacks empirical studies. Finally, the PC performed a higher TTV than the IC. Given that both groups trained at a similar relative intensity (i.e., 8–12RM) and the rate of progression was similar, the higher TTV in the PC was likely due to the greater absolute weight lifted in the preacher curl exercise compared to the incline curl. Based on this, one might argue that the higher VTT may confer an advantage to the PC over the IC group. Given that we saw specificity in strength gains, the results of our study do not support this hypothesis. The same is true for the muscle growth outcome; there did not appear to be a difference between high- and low-load resistance training [35]. Furthermore, the absolute weight lifted difference between the two exercises is likely an inherent feature of the task (e.g., stability and muscle length differences). Thus, our study maintains high ecological validity by comparing the two exercises with the inherent characteristics associated with each movement (e.g., lifting more or less absolute weight).

Conclusions

Our findings suggest that the muscle adapts differently depending on the resistance exercise mechanical characteristics and the length at which the target muscle is being trained. More specifically, the incline biceps curl elicits slightly greater muscle growth of the proximal elbow flexors, while the preacher biceps curl induces the greater development of the distal elbow flexors after 8 weeks of regimented resistance training. Regarding muscle strength, exercise selection influences the magnitude of adaptations; the changes seem to follow the principle of specificity. The incline biceps curl training elicits greater strength gains in the incline biceps curl exercise while the preacher biceps curl training elicits greater strength gains in the preacher biceps curl exercise. Our findings can be used to guide the exercise selection in resistance training programs aiming to optimize muscle hypertrophy and strength gains. Given the apparent complementarity of the two exercises induced adaptations, it is reasonable to recommend that strength and conditioning professionals and practitioners consider both biceps curl resistance exercises, particularly when the objective is to optimize the hypertrophy of the different muscle sites and strength improvements in both tasks. Of note, direct evidence of the effects of combining the two resistance exercises is needed.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgement

We would like to express thanks to all the participants for their engagement in this study, the Coordination of Improvement of Higher Education Personnel (CAPES/Brazil) for the scholarship conferred to WK, BC, GK, FL, NS, AP, IT, JF, and LA, and the National Council of Technological and Scientific Development (CNPq/Brazil) for the grants conceded to ESC. The authors also would like to extend our heartfelt thanks to the following research assistants, without whom this study could not have taken place: Felipe Gomes de Paula.

Conflict of Interest

The authors declare that they have no conflicts of interest.

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